Moving Megawatts: How electricity market zone splits impact charging infrastructure planning for long-haul electric trucks

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***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

A. Motivation

Across Europe, there is an ongoing discussion on whether large, uniform electricity bidding zones should be reconfigured toward smaller zones that better reflect local pricing signals [3], [7]. This debate is driven by persistent grid congestion that arises when uniform wholesale prices fail to reflect the spatial mismatch between generation and demand. As renewable generation—particularly wind power—concentrates in specific regions, the mismatch between generation and demand creates systematic congestion patterns that uniform pricing cannot resolve, leading to costly redispatch interventions and inefficient investment signals.

One of the most prominent instances of this debate is the potential splitting of the German electricity market zone [1]–[3], [7]. Germany currently operates as a single bidding zone despite significant structural congestion along its North–South axis: wind-rich northern regions generate surplus electricity that must be transmitted to demand-heavy southern industrial centers [1]–[3], [8]–[10].

Simulation studies have shown that a German North–South split reduces redispatch but creates significant distributional effects on electricity prices, generation investment, and welfare [1]–[3], [7]. Beyond the electricity sector, zone reconfigurations cascade into end-use sectors: they alter the feasibility of electrification in heating, transport, and industry [9], [10], change investment viability of storage infrastructure [8], and affect demand differently across economic sectors [11]. A consumer sector rarely discussed in this context is the transport sector. Charging costs—largely determined by local electricity prices—significantly affect the cost competitiveness of battery-electric trucks [12], [14], yet the effect of price differences across electricity market zones on the spatial allocation of charging demand has barely been studied, with the exception of Pagnier et al. [13] who find a relevant link between zonal electricity prices and optimal truck electrification in Norway.

An added dimension in the transport sector, compared with stationary consumers such as industry or heating, is that transport consumers are mobile: long-haul trucks travel across multiple regions and, for international freight, across national borders and electricity market zones. This mobility creates an opportunity—when charging is optimized over the route—to exploit spatial price differentials by shifting charging demand toward regions with lower electricity costs, thereby improving cost competitiveness. It also means that a zone split in one country (here, Germany) can redistribute charging loads not only within that country but along the entire transit corridor.

B. Research Objective

Research questions need to be adapted. This leads to the following research question: **How does a splitting of the German electricity market zone affect (1) CO_2 emissions from freight transport on German highways, (2) the spatial distribution of truck charging demand between Northern and Southern Germany, and (3) the risk of stranded charging infrastructure when the split is introduced at different points in time?**

We address this by simulating a market split via expected electricity price levels under different conditions of the split (spread magnitude and timing), deriving location-specific charging costs, and modeling the long-term demand response with a linear program for cost-optimal multi-year fleet turnover and charging demand allocation along origin-destination truck flows on the Scandinavian-Mediterranean corridor.

II. Methodology and Data

A. German market zone splitting

We simulate the impact of a German electricity market zone split by constructing a scenario matrix that varies two dimensions: (i) the *magnitude* of the North–South electricity price spread, and (ii) the *timing* of when the zone split takes effect. The scenario matrix is summarized in Table I.

1) Scenario design: A single *baseline* scenario assumes a unified German price zone throughout the modeling horizon (2020–2050), corresponding to the current market design. Six zone-split scenarios are constructed by combining three

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spread levels with two implementation timings (from 2032 and from 2040).

Price spread levels. The price differential between the North and South German zones is applied as a symmetric, relative adjustment to the baseline wholesale electricity price at each node:

- **Low (5%):** North -2.5%, South $+2.5\%$ — conservative, based on early estimates by Trepper et al. [1]
- **Mid (10%):** North -5%, South $+5\%$ — central estimate, consistent with Knörr et al. [3] and Fraunholz et al. [2]
- **High (20%):** North -10%, South $+10\%$ — upper bound, reflecting early-period spreads in Fraunholz et al. [2]

Using a relative (percentage-based) differential ensures that the absolute price spread scales with the price level: larger in early years when wholesale prices are higher, and smaller as prices decline with renewable energy expansion.[1]

Implementation timing. The zone split takes effect in one of two years:

- **From 2032:** prices remain unified until 2031; split from 2032 onward
- **From 2040:** prices remain unified until 2039; split from 2040 onward

Before the split year, all German nodes retain the baseline (unified) electricity price. This dimension captures the policy uncertainty around *when* such a reform might be implemented, and whether early versus late splitting has different effects on the already-established truck fleet and charging infrastructure.

Zone definition. Germany is divided into two zones following the TSO boundary convention used in Trepper et al. [1] and Fraunholz et al. [2]:

- **North** (50Hertz + TenneT-North): wind-rich regions with lower marginal generation costs (Lower Saxony, Schleswig-Holstein, Mecklenburg-Vorpommern, Brandenburg, Berlin, Saxony, Saxony-Anhalt, Thuringia, Bremen, Hamburg)
- **South** (Amprion + TransnetBW + Bavaria): demand-heavy regions with higher marginal costs (NRW, Hesse, Rhineland-Palatinate, Saarland, Baden-Württemberg, Bavaria)

2) Electricity prices and charging costs: Baseline wholesale electricity prices are taken from the **Go RES** scenario of GENeSYS-MOD [5], which assumes ambitious renewable energy expansion in line with the EU's Green Deal targets. This scenario was selected because it represents a central pathway with high renewable penetration—precisely the conditions under which North–South price differentials in Germany are expected to be most pronounced due to the geographic concentration of wind power in the North. Go RES

[1]For the High (20%) scenario, the resulting fast-charging price spread (including CPO markup) between North and South Germany is approximately 5.2 ct/kWh in 2030 and 2.9 ct/kWh in 2040; for Mid (10%) it is 2.6 and 1.5 ct/kWh; for Low (5%) it is 1.3 and 0.7 ct/kWh, respectively. For a BEV truck consuming $\sim$1.2 kWh/km over a typical $\sim$600 km German transit segment, the High scenario translates to a cost difference of roughly EUR 37 per trip in 2030 and EUR 21 in 2040 between charging entirely in the North versus entirely in the South.

TABLE I
SCENARIO MATRIX: 1 BASELINE + 6 ZONE-SPLIT SCENARIOS

	Split timing	
Spread	From 2032	From 2040
Baseline (0%)	DE_unified (reference)	
Low (5%)	L-2032	L-2040
Mid (10%)	M-2032	M-2040
High (20%)	H-2032	H-2040

provides node-level annual wholesale electricity prices for 2020–2050 across all corridor countries (NO, FI, SE, DK, DE, AT, IT). For zone-split scenarios, the German prices are adjusted as described above: for each German node, the wholesale price is multiplied by $(1-\delta/2)$ in the North zone or $(1+\delta/2)$ in the South zone, where δ is the relative spread (5%, 10%, or 20%). This adjustment is only applied from the split year onward; earlier years retain the unified price.

Local charging costs are composed of: wholesale electricity price, network charges (grid fees), CPO markup, and charging infrastructure investment costs. Describe each component and how they vary spatially. Include map showing two snapshots of charging cost landscape.

Fig. 1 shows the resulting electricity charging prices (including CPO markup) for all four spread levels with the earliest timing (from 2032). The North–South differential is clearly visible for the Mid and High spread cases, while non-German countries remain unaffected.

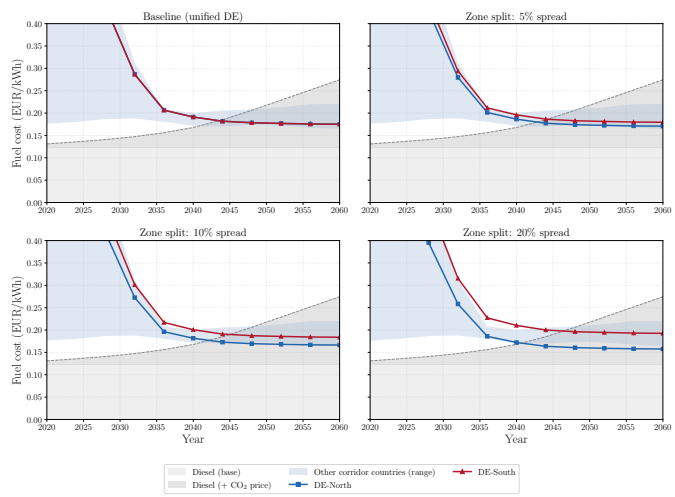

Fig. 1. Fast-charging electricity prices by country and German zone for the four spread levels (split from 2032). Gray areas show diesel cost (base and including CO_2 price). DE-North (squares) and DE-South (triangles) diverge with increasing spread.

B. Cost-optimization of fleet turnover and charging demand allocation

The spatial allocation of charging demand and fleet composition is determined by a linear program that minimizes total

discounted system cost over the horizon 2020–2050 with a 4-year time step. The model, TransComp [16], is formulated as a network design problem and determines, for each OD pair and period, the cost-optimal mix of ICEV and BEV trucks, the spatial allocation of charging energy along routes, and the required charging infrastructure investments. Table II summarizes the key notation.

TABLE II
KEY NOTATION

Sets	
$y \in \mathcal{Y}$	modeled years
$r \in \mathcal{R}$	origin-destination pairs
$t \in \mathcal{T}$	drivetrain technologies (ICEV, BEV)
$g \in \mathcal{G}$	vehicle generation (purchase year)
$e \in \mathbf{E}_r$	geographic locations along route r
Decision variables	
f_{yrtg}	transport volume [t/a]
h_{yrtg}, h^+_{yrtg}	fleet stock; new purchases
s_{yrtge}	charged energy at location e [kWh/a]
soc_{yrtge}	state of charge at e [kWh]
q^+_{ye}	added charging infra. capacity [kW]
Key parameters	
F_{yr}	freight demand [t/a]
D^{spec}_{tg}	specific energy consumption [kWh/km]
Q^{batt}_{tg}	battery/tank capacity [kWh]
$p^{\text{fuel}}_{te}(y)$	location-specific fuel price [EUR/kWh]
$\pi^{\text{CO}_2}_e(y)$	carbon price [EUR/tCO_2]

1) Objective function: The objective minimizes the net present value of all costs:

$$\min_{\mathbf{x}} \; z = \sum_{y \in \mathcal{Y}} d_y \, \Delta \Big(C^{\text{veh}}_y + C^{\text{fuel}}_y + C^{\text{toll}}_y + C^{\text{travel}}_y + C^{\text{charg}}_y \Big) \tag{1}$$

where $d_y = (1+\rho)^{-(y-y_0)}$ is the discount factor and Δ is the period length (4 years). Vehicle costs C^{veh} include CAPEX of new purchases and age-dependent maintenance. Fuel costs C^{fuel} include location-specific energy prices and a carbon surcharge for fossil fuels: $\sum_{r,t,g,e} s_{yrtge} \cdot (p^{\text{fuel}}_{te}(y) + \varepsilon_t \cdot \pi^{\text{CO}_2}_e(y))$. Toll costs C^{toll} are technology-differentiated distance-based road charges. Travel time costs C^{travel} monetize only the *deviation* from baseline travel time (driving + mandatory breaks) using a value of time, so that charging during mandatory driver rest incurs no additional cost. Charging infrastructure costs C^{charg} cover CAPEX, O&M, and grid connection charges for deployed capacity.

2) Key constraints: **Demand coverage.** All freight demand must be served:

$$\sum_{t,g} f_{yrtg} = F_{yr} \quad \forall\, y, r \tag{2}$$

Fleet turnover. The vehicle stock is tracked by generation through a stock-flow balance of existing vehicles, retirements (after fixed lifetime Λ_{tg}), and new purchases, capturing the fleet inertia that constrains the pace of technology transition.

State of charge. The battery state of charge is tracked spatially along each route:

$$soc_{yrtge_i} = soc_{yrtge_{i-1}} + s_{yrtge_i} - D^{\text{spec}}_{tg} \cdot L_{e_{i-1},i} \cdot \frac{f_{yrtg}}{W_{tg}} \tag{3}$$

with initial SOC set to $SOC^{init} \cdot Q^{\text{batt}}_{tg}$ and a minimum final SOC constraint. The SOC must remain within $(0, Q^{\text{batt}}_{tg})$ at all intermediate nodes, preventing stranding and ensuring feasibility of BEV operation. This constraint is the key mechanism through which electricity prices affect spatial charging allocation: the optimizer distributes charged energy s along the route to minimize fuel cost subject to the battery capacity envelope.

Charging infrastructure. Installed capacity (initial + cumulative expansion) must meet peak demand at each location:

$$Q^{\text{charg}}_e + \sum_{y' \le y} q^+_{y'e} \;\ge\; \gamma \sum_{r \in \mathcal{R}_e} \sum_{t,g} s_{yrtge} \quad \forall\, y, e \tag{4}$$

where γ converts annual energy to peak power demand. Infrastructure, once built, is irreversible—creating path dependencies when the zone split is introduced at different times.

C. Traffic flow data

The model operates on origin-destination (OD) freight flows at the NUTS-2 level along the Scandinavian-Mediterranean TEN-T corridor, covering Norway, Finland, Sweden, Denmark, Germany, Austria, and Italy. Freight volumes are derived from the ETISplus European freight transport database [15], which provides bilateral road freight tonnages between NUTS-2 regions. Routes are computed as shortest-path sequences of NUTS-2 regions over the highway network, and mandatory driver rest stops are allocated along each route following EU Regulation 561/2006 (4.5 h maximum continuous driving time) [16].

Two route categories are included:

1) **International routes passing through Germany:** cross-border OD pairs whose path transits through at least one German NUTS-2 region (e.g., IT↔DE, SE↔DE, IT↔SE via DE). International routes that bypass Germany entirely (e.g., AT↔IT, DK↔NO via SE) are excluded since they are unaffected by the German zone split and would produce identical results across all scenarios.
2) **German domestic N–S routes:** OD pairs where both origin and destination are in Germany but in *different* electricity market zones (North↔South). These routes are essential for the zone-splitting analysis because trucks traveling within Germany across the zone boundary can reallocate charging between cheaper (North) and more expensive (South) regions.

Routes are filtered to include only those with a minimum path length of 360 km (corresponding to ∼4.5 hours of driving at 80 km/h, i.e., one mandatory driver rest interval), a minimum freight volume of 500 tonnes/year in the base year, and transit through at least one German region. After filtering, the dataset contains approximately 1,787 international OD

pairs transiting through Germany and 83 German cross-zone domestic OD pairs, for a total of ∼1,870 routes. The 249 international routes bypassing Germany (e.g., AT↔IT, Scandinavian intra-Nordic) are excluded to reduce computational cost without affecting results.

III. Results

A. Route-level electrification shifts

While the aggregate BEV fleet share remains nearly identical across all scenarios (∼43.7% at 2040), the zone split fundamentally reshapes *which* routes electrify. Fig. 2 shows the distribution of route-level BEV share changes between the High (20%) from-2032 scenario and the baseline at 2040, disaggregated by corridor type. Routes are classified as *North only* if their path passes exclusively through DE-North NUTS-2 regions, *South only* if exclusively through DE-South, and *cross-zonal* if through both.

North-only routes gain on average +21 pp in BEV share, with many routes shifting from ∼15% to 100% BEV—corresponding to approximately 1,230 additional tonnes transported by BEV trucks on these corridors. Conversely, South-only routes lose on average −18 pp, with ∼2,980 tonnes shifting back from BEV to ICEV. Cross-zonal routes show mixed effects (mean +4 pp), with 245 routes gaining and 100 losing more than 5 pp BEV share.

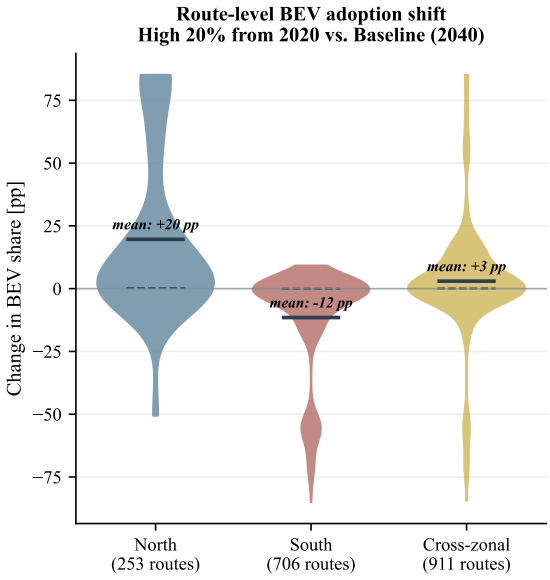

Fig. 2. Distribution of route-level BEV share changes (High 20%, from 2032 vs. baseline) at 2040 by corridor type. Solid line: mean; dashed line: median. North-only routes gain BEV share while South-only routes lose it, despite identical aggregate fleet composition.

B. Spatial redistribution of charging demand

IV. Conclusion and Future Work

The zone split does not change the aggregate BEV fleet size but fundamentally reshapes *where* charging demand arises, creating regional winners and losers along the Scandinavian–Mediterranean freight corridor. A promising extension is a **rolling-horizon** formulation: instead of a single-shot optimization over the full planning horizon, infrastructure investment decisions would be made with limited foresight. This would capture the risk of building charging capacity under one market-zone configuration (unified) that becomes underutilized if the zone split occurs later—providing a more realistic estimate of stranded infrastructure costs.

Acknowledgment

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References

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APPENDIX

Table III summarizes the key cost and technical parameters for the two drivetrain technologies considered in this study.

TABLE III
VEHICLE COST AND TECHNICAL ASSUMPTIONS

Parameter	**Unit**	**ICEV**	**BEV**
Capital cost (CAPEX)			
2020	kEUR	169	550
2030	kEUR	192	239
2040	kEUR	192	218
Operating cost			
Maintenance	kEUR/a	20.7	14.4
Toll	EUR/km	0.100	0.036
Energy			
Spec. consumption 2020	kWh/km	3.64	2.06
Spec. consumption 2040	kWh/km	2.40	1.29
Tank / battery capacity	kWh	5,529	1,187
Diesel price	EUR/kWh	0.123 (constant)	
CO_2 emission factor	gCO_2/kWh	266	0
Other			
Payload capacity	t	13.6	10.9
Vehicle lifetime	a	12	12
Annual mileage	km/a	136,750	
Discount rate	–	5%	
Value of time	EUR/h	30	